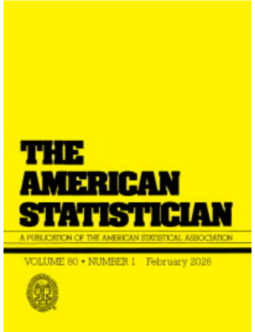

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Seeing Different Stories: Human Variability in Statistical Assumption Checking and Model Selection

Gavin Tovar & Sarah Emerson

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Gavin Tovar* and Sarah Emerson

Department of Statistics, Oregon State University

*tovarg@oregonstate.edu

Abstract

Inspired by recommendations commonly encountered in introductory statistics classes to perform model checking and make modeling decisions based on visual inspections of diagnostic plots, we implemented and administered a survey on interpretation of a variety of diagnostic plots across multiple settings. We summarize the patterns in the answers provided by more than 300 undergraduate and graduate students enrolled in statistic courses via a mixed effects model exploring the impact of different data-generating settings and various plot features. Notably, we find that the students surveyed here are comparable to random chance at making the correct decisions on the basis of the plots they were shown. The implications of this uncertainty and the lack of reproducibility introduced by the model checking/model selection process are explored in depth in one of the settings presented to demonstrate the consequences of using subjective and frequently sub-optimal human judgments to make modeling choices.

Keywords: Diagnostic plots; post-selection inference; model-checking; regression diagnostics; reproducibility

1 Introduction

Statistical models rely on a variety of assumptions—such as linearity, homoscedasticity, independence, and normality—to ensure valid inference. Some of these assumptions are primarily relevant to regression models, but many are also highlighted in the derivation and teaching of various other procedures. Diagnosing potential violations of these assumptions is a standard and ubiquitously taught part of statistical modeling workflows. Nearly all introductory statistics curricula, textbooks, and software tools emphasize the importance of checking these assumptions through diagnostic plots. Indeed, a review of widely used texts—including *Statistics* (Freedman et al., 2007), *Introduction to Statistical Learning (ISL)* (James et al., 2013), *Modern Statistics with R* (Thulin, 2024), *The Statistical Sleuth* (Ramsey and Schafer, 2012), *OpenIntro Statistics* (Diez et al., 2019), *Introduction to Modern Statistics (2e)* (Çetinkaya-Rundel and Hardin, 2024), and *Applied Linear Regression (ALR)* (Weisberg, 2005)—reveals a strong consensus around using graphical methods such as residual plots, quantile-quantile (Q-Q) plots, and case influence statistics diagnostics as the primary tools for assumption checking.

These graphical methods are recommended for nearly every major assumption and data quality concern of regression and other statistical procedures:

• For assessing **normality**, *ALR*, *Modern Statistics with R*, and *The Statistical Sleuth* consistently recommend normal probability plots (Q-Q plots). *Introduction to Modern Statistics (2e)* highlights residual symmetry and the careful handling of outliers as indicators of normality violations.

• For **linearity**, residuals vs. fitted value plots are ubiquitous. *ISL* and *Modern Statistics with R* encourage the use of LOESS smoothers to detect curvature. When nonlinearity is evident, texts suggest transformations (e.g., $\log$, $\sqrt{\cdot}$) or polynomial terms. *Statistical Sleuth* offers a visual taxonomy of nonlinearity cases, recommending transformations or quadratic regression depending on the resulting plots.

• Regarding **constant variance (homoscedasticity)**, most texts—including *ISL*, *ALR*, and *Modern Statistics with R*—emphasize scale-location plots or residual plots. Remedies include transformations of the response (e.g., $\log(Y)$) or weighted least squares. Formal tests like the Breusch–Pagan test are occasionally mentioned, but generally subordinated to visual inspection.

• For **independence**, few graphical diagnostics are recommended. Instead, textbooks like *ISL* and *Introduction to Modern Statistics* emphasize study design and caution against applying regression to time-series or spatial data without appropriate models. *Statistical Sleuth* suggests plotting residuals in temporal order to check for serial correlation.

• To identify **outliers** and **influential points**, texts agree on the utility of studentized residuals, Cook's distance, and leverage plots. *ISL* notes a studentized residual greater than ± 3 as a rule-of-thumb threshold, while *Modern Statistics with R* advocates for plotting Cook's distance vs. leverage to locate points that may disproportionately impact model fit. *ALR* provides a formal test for outliers via a "mean shift" model.

• **Multicollinearity**, though not detectable through residual plots, is frequently addressed via scatterplot matrices and correlation diagnostics. *ISL* and *Modern Statistics with R* recommend removing or combining correlated predictors, or applying regularized regression techniques. *Statistical Sleuth* adds that parsimony improves inference precision and provides multi-step model selection strategies.

• For **transformations**, many of the books mentioned converge on the use of $\log$, $\sqrt{\cdot}$, or $1/x$ functions to correct for violations of normality, non-linearity, or unequal variance. *Modern Statistics with R* and *Statistical Sleuth* also introduce Box–Cox transformations and robust regression as alternatives.

Taken together, these recommendations underscore how graphical diagnostics have become entrenched in statistical pedagogy, not only as tools for model assessment, but also as vehicles for understanding regression assumptions. However, the educational pervasiveness of these plots may conceal their limitations, especially when they are used uncritically or interpreted without awareness of human perceptual biases.

The path of selecting modeling and data analysis strategies is full of decision points that are often poorly understood and documented, and generally approached very subjectively. Student difficulty with making modeling decisions is a common hurdle (Allen et al., 2016), and the hope of finding answers in plots is enticing. While some advocate supplementing visual checks with formal statistical tests (e.g., Shapiro–Wilk, Breusch–Pagan), these approaches are themselves fraught with challenges like their sensitivity to sample size and encouragement of dichotomous reasoning (Shatz, 2024). In practice, these tests may flag minor violations that are not practically meaningful or miss meaningful deviations from assumptions due to low power, and in all settings risk inflating post-model selection bias and compounding the multiple testing problem. As a result of these challenges with formal assumption tests, diagnostic plots have become a widely recommended alternative, claiming a more nuanced, visual approach to assessing assumptions and guiding model refinement (Fox, 1991; Reinhart, 2025).

Despite the widespread use and endorsement of diagnostic plots, a growing body of research highlights that individuals—even those with formal training—often struggle to correctly interpret them. For example, Ravinder et al. (2019) found that undergraduate students frequently misinterpreted residual plots and often failed to draw appropriate conclusions about regression assumptions. The difficulty in interpreting plots is not merely a pedagogical concern; it reflects deep-seated cognitive and perceptual biases. Ciccione and Dehaene (2021) demonstrated that even brief exposure to scatterplots allows viewers to detect general trends, but with systematic distortions—such as overestimating slopes—suggesting perceptual heuristics inconsistent with classical regression theory. Similar biases have been observed across a range of contexts: individuals tend to misjudge and differentially estimate correlations based on axis scaling (Cleveland et al., 1982), anchor their magnitude estimates to axis limits (Bradley et al., 2025), and exhibit task-dependent visual—spatial biases in scatterplot interpretation (Divis et al., 2023).

These findings echo longstanding research in graph comprehension, which shows that graph format, data density, and viewer expectations can all influence interpretation (Lauer and Post, 1989; Kosslyn, 1985). Some similar investigations into how preconceived biases affect interpretation of plots is nicely detailed in Patitsas et al. (2019), finding that different viewers report different interpretations of the bimodality of a histogram, particularly if they are primed to detect such bimodality. While tools like the `regressinator` (Reinhart, 2025) offer simulation-based teaching tools to reinforce statistical intuition—such as sampling distributions and assumption violations—certain recommendations, such as the use of "visual hypothesis tests" through simulated diagnostic lineups (Buja, Cook, Hofmann, Lawrence, Lee, Swayne and Wickham, 2009), raise concerns. These strategies blur the line between pedagogical illustration and inferential rigor. While lineups may help students recognize extreme patterns, they implicitly encourage an informal testing framework that does not account for the complexities of multiple comparisons or subjective interpretation. Moreover, though useful for teaching, such tools may not generalize well to research practice without introducing bias.

Existing studies, including the nicely designed and investigated survey similar to this one in Ravinder et al. (2019), often assess viewer accuracy in simplified or narrow contexts—such as binary choices about trend direction or normality. Moreover, most prior work has not systematically addressed how data features, like sample size or the severity of assumption violations, interact with visual judgments across diverse diagnostic tasks. Nor have they accounted for between-dataset variability, which may be just as important as between-participant variability in shaping interpretations. We explore some of these additional dimensions of the subjectivity and variability of plot interpretations through an online survey administered to more than 400 students in a wide range of statistics classes. To quantitatively compare the effects of these different factors in practitioners' ability to correctly interpret plots, we then construct an overarching model of the results of this survey with factors for question type, specific dataset, specific study participant, and plot and dataset features.

1.1 Contribution of This Study

This study contributes to the growing literature on statistical cognition by empirically investigating how individuals assess diagnostic plots across a variety of modeling contexts and plot/question types. We investigated this broad question through the implementation of a Qualtrics survey offered to all students, both undergraduate and graduate, enrolled in Statistics classes at Oregon State University during Spring 2025. The survey presented participants with a series of plots and asked them to select from multiple choice answers to interpret the plots. Our factorial survey design crosses assumption validity and sample size across both fixed and randomly generated datasets, allowing us to examine:

• How interpretation varies across individuals and across datasets;

• The influence of visual aids like smoothing curves (e.g., LOESS) on judgment;

• Whether key data features (such as sample size and assumption validity) systematically affect interpretation.

Our results highlight that variability across datasets can exceed variability across respondents in the data-generating settings considered here, suggesting that interpretation is not only subjective but also highly context-dependent. We further emphasize that the notion of "truth" in assumption-checking should not be viewed as binary. Instead, violations lie along a continuum, and interpretation accuracy may reasonably scale with the degree of deviation—though we do not formally test this gradient in our analyses.

We demonstrate that assumption-checking is not a mechanical procedure, but a cognitively and visually complex process, shaped by both human variability and data structure. These findings also underscore the potential risks of relying on visual diagnostics in high-stakes modeling workflows, especially when interpretive uncertainty is underappreciated or ignored, and reproducible analyses are important.

2 Survey Implementation

2.1 Survey Design

To investigate how individuals interpret statistical diagnostic plots and make modeling decisions, we developed a survey targeting participants with varying levels of statistical training. The primary goal was to assess how visual information shapes decisions about model assumptions, and to quantify the variability in such interpretations. This variability is a key contributor to post-model selection bias—when modeling decisions differ across analysts, resulting in downstream inference that is drawn from a mixture of models.

The survey, implemented with IRB approval, was administered anonymously via Qualtrics to undergraduate and graduate students enrolled in statistics courses. Subjects were recruited from all statistics courses being taught at Oregon State University during Spring term 2025 via emails to class lists. To encourage participation, respondents were offered entry into a gift card raffle upon completion.

The instrument included 25 questions for each participant, each designed to probe participants— evaluations of diagnostic plots related to transformations, regression assumptions (e.g., normality, linearity, and homoscedasticity), outlier and influence diagnostics, and variable selection strategies. Two of the question types were duplicated with and without a LOESS curve to assess whether smoothed visual cues affected interpretation.

To systematically evaluate how data features influence judgment, each question type (with the exception of a forest plot interpretation) was embedded in a controlled experimental design. Each question type was asked under either four or eight settings, consisting of low or high sample size ($n = 30$ or $n = 100$, respectively) crossed with different settings for the assumption or data feature in question (assumptions met or not met, for question types 1-7 and 9-11; question types 8, 12, and 13 included eight variations each to cover more complex settings related to variable transformation, two-sample testing strategies, and meta-analysis contexts). For the first twelve question types, each participant viewed two datasets: one of four or eight fixed datasets depending on the number of settings for that question type, and one randomly selected from a larger population of datasets from the same data-generating mechanism. An example set of survey questions is provided in Appendix A.1. Examples of the plots for each question under each of the different combinations of sample size and assumption validity are given in the Supplementary Materials.

By incorporating repeated question formats (e.g., with and without LOESS smoothing), we were also able to quantify how subtle visual modifications impact the interpretation of statistical evidence. Specifically, question types 4 and 5, as well as 6 and 7, were later combined in the analysis to isolate the effect of the LOESS overlay on participants' responses. This factorial design, summarized in Figure 1, enabled us to investigate the effects of sample size and assumption validity on decision-making performance, allowing for a detailed analysis of response patterns under varying inferential conditions.

2.2 Data Quality and Cleaning

A total of 421 respondents participated in the survey, but a subset of 44 participants left all questions unanswered, and another 37 answered entirely "yes" or "no". Removing the respondents who gave all "yes" or all "no" responses had very little impact on the resulting data summaries, but the results presented here exclude these subjects under the supposition that they were not actively engaging with the survey.

The majority of respondents were undergraduates (45 graduate students, 7 "other", and 369 undergraduate students), with an average of 2.94 statistics classes taken per respondent. A total of 263 respondents completed the entire survey, and the average number of questions answered per respondent was 17.06 (including the 44 who answered zero questions). As noted by a reviewer, the study could have been strengthened by incorporating attention-check questions to more directly assess participant engagement.

Since the sample was drawn from a single institution, institutional bias is possible, and replication at additional universities is encouraged. More broadly, there were limitations in the scope and implementation of this survey that may reduce the immediate generalizability of our findings to a wider population of students or statistical practitioners. Nonetheless, based on our experience and prior literature in related domains, we suspect that many of the observed patterns in interpretation differences would likely persist across other populations and settings. Our primary goal in this investigation was to explore—and make initial attempts to quantify—how divergent viewers—interpretations of the same plots can be.

3 Survey Results

3.1 High-Level Observations

Figure A.2 in Appendix A.3 shows the proportion of correct responses for each question, grouped by the number of statistics courses respondents had completed. One might expect—or at least hope—that more statistical training would correspond to greater accuracy in identifying correct assumption settings. While this pattern emerges in a few cases, it is far from consistent. Notably, respondents with 0–3 statistics courses often performed within 5–15 percentage points of those with 15 or more courses.

Figure A.3 compares accuracy between low and high sample sizes across various assumption settings, aggregated over all questions. Contrary to expectation, larger samples did not produce higher accuracy. The common intuition—that assumption violations should be easier to detect in larger datasets—finds little support in these results.

Figure A.4 presents the overall distribution of correct responses across question types. The histogram's mass centers around 0.50, suggesting that identifying assumption violations from diagnostic plots was, on average, close to random guessing.

Figure A.5 provides a question-by-question breakdown of accuracy, stratified by both sample size and assumption validity. Once again, there is little evidence that larger samples aid in evaluating these diagnostic plots, even when controlling for the type of assumption under consideration.

Finally, Table 9 summarizes participant responses for all question types and experimental conditions. Each cell reports choices stratified by sample size (low or high) and whether the relevant assumption held. Rows indicate response options—usually binary (Yes/No), but in some cases involving multiple modeling strategies such as transformation types or hypothesis tests.

3.1.1 Question Breakdown

A brief summary of each question is given below, along with high-level patterns noticed in the responses. Table 1 summarizes the proportions and percentages of correct responses for each question broken down by question setting (high/low sample sizes, and validity of assumptions).

Question 1 (Figure 1) assessed visual evaluation of normality using histograms and Q–Q plots. Normality is a foundational assumption in many statistical analyses, and it's common for researchers to visually assess this when deciding on modeling strategies. Despite the centrality of normality in statistical inference, participants tended to affirm the assumption even when it was violated. As shown in Table 9, responses were balanced only under low sample size with assumption violation. Figure A.5 further illustrates that correctness dropped below 20% when assumptions were violated at high sample size, but approached 80% when assumptions were satisfied.

Question 2 (Figure 2) focused on interpreting log transformations. Many practitioners routinely use log transformations to address skewness or non-linearity. In this context, correct inference about the median on the original scale depends on the log-transformed data being approximately symmetric. Table 9 and Figure A.5 show a near-even response split, with performance below random chance under low sample sizes and only marginally better under high sample sizes.

Question 3 (Figure 3) addressed decisions around variable selection in the presence of multicollinearity. Assessing and managing correlated predictors is a routine task in regression modeling. Participant responses lacked consistency, appearing nearly random across all conditions. Figure A.5 suggests that correctness was worse than random guessing in high sample size scenarios when assumptions were violated.

Questions 4 and 5 (Figures 4 and 5) assessed the linearity assumption in regression, without and with the inclusion of a LOESS curve. Linearity is essential for unbiased parameter estimation, and diagnostic plots are standard tools for checking this. While diagnostic plots are common tools for this assumption in particular, responses remained inconsistent. The LOESS curve nudged participants toward rejecting linearity more often, likely reflecting overinterpretation of the smoothed fit. Figure A.5 shows that for Question 4, participants more accurately detected violations than confirmations. For Question 5, the LOESS curve improved correct identification when linearity held but hindered it when it did not.

Questions 6 and 7 (Figures 6 and 7) examined constant variance (homoscedasticity) through residual plots, again without and with LOESS. While some responses matched the underlying variance structure, there was still substantial disagreement, especially in low sample size settings. Unlike the linearity questions, the LOESS overlay had minimal effect (Table 9), though Figure A.5 shows slightly reduced correctness in most conditions.

Question 8 (Figure 8) asked participants to choose the most appropriate transformation (on X, Y, both, or neither) for regression assumptions. The most frequent choice—"Both"—persisted across all conditions, even when unwarranted. Table 9 shows little sensitivity to assumption validity, and Figure A.5 indicates "Both" yielded about 50% correctness, likely due to the large selection, while other options performed notably worse.

Question 9 (Figure 9) involved detecting high-leverage points using leverage plots. Participants predominantly favored further investigation or removal, particularly when assumptions were violated or sample sizes were large. Figure A.5 shows this caution yielded high correctness under violation, but low correctness when assumptions held.

Question 10 (Figure 10) focused on Cook's distance, a common influence statistic. Compared to leverage, participants were even more inclined to flag points—"Yes" responses exceeded "No" responses across all conditions. Figure A.5 echoes the pattern from Question 9: eagerness to act produced correctness when issues existed, but poor performance when assumptions were met.

Question 11 (Figure 11) used standardized residuals to identify outliers. Responses were more restrained compared to leverage and Cook's distance, with a modest margin of correct over incorrect answers. Figure A.5 shows relatively strong performance across all conditions—correctness ranged from 56% to 68%, around the best performance among all questions.

Question 12 (Figure 12) tested selection of statistical tests under varying assumptions. Despite normality violations, participants overwhelmingly favored *t*-tests over non-parametric alternatives. Choices between equal-variance and unequal-variance *t*-tests appeared nearly random. The Wilcoxon test was slightly preferred over the permutation test, though both were rarely selected. Figure A.5 shows accurate identification of equal variance when it held, but only 50% correctness when it did not.

Question 13 (Figure 13) introduced forest plots—less familiar to many participants—to assess treatment effects from study-level confidence intervals. Despite limited exposure, and with only a brief explanatory note provided, most participants concluded a treatment effect existed, regardless of the actual scenario. Surprisingly, Figure A.5 shows better performance under low sample sizes.

3.2 Result-Evaluating Model

After removing data from participants whose responses indicated lack of engagement with the survey, we were left with a total of 340 participants who collectively answered 7099 questions. For each of these questions, we ascertained what the correct answer was based on the data generating mechanism (details on the correct answers for each question type and setting are given in Appendix A.2, Table 8).

To obtain a high level summary of the impact of question type, sample size, loess presence, statistical education, and the validity of data-generating assumptions—while accounting for variability across participants and datasets—we fit the following mixed-effects model to the binary indicator of correct answer for each of the 7099 answered questions in the cleaned dataset:

$$y_{ijklmp} = \mu + \alpha_i + \beta_j + \gamma_k + \delta_\ell + \tau_m + \eta_{p(k,j(i))} + \theta x_{ijklmp} + \epsilon_{ijklmp}$$

where

- $y_{ijklmp} =$ correctness (binary) for question *i*, setting *j* (within *i*), sample size *k*, loess status ℓ, participant *m*, and dataset *p* (within *k* and within *j* (within *i*))

- $\mu =$ overall mean

- $\alpha_i =$ fixed effect of i^{th} question type

  - $\sum_i \alpha_i = 0, \quad i \in \{1,2,3,4,6,8,9,10,11,12,13\}$

- $\beta_{j(i)} =$ fixed effect for j^{th} assumption setting nested in i^{th} question type

  - $\sum_{j(i)=1}^{n_i} \beta_{j(i)} = 0, \quad j(i) = 1, \ldots, n_i$

- $\gamma_k =$ fixed effect for k^{th} sample size

  - $\sum_{k=1}^{2} \gamma_k = 0, \quad k = 1, 2$

-

$\delta_\ell =$ fixed effect for ℓ^{th} loess setting

- $\sum_{\ell=1}^{2} \delta_\ell = 0, \quad \ell = 1, 2$

•

$\tau_m =$ random effect for m^{th} participant

- $\tau_m \overset{\text{iid}}{\sim} N(0, \sigma_\tau^2), \quad m = 1, \ldots, 340$

•

$\eta_{p(k,j(i))} =$ random effect of p^{th} dataset nested in k^{th} sample size and the j^{th} setting which is also nested in i^{th} question type

- $\eta_{p(k,j(i))} \overset{\text{iid}}{\sim} N(0, \sigma_\eta^2), \quad p(k, j(i)) = 1, \ldots, q_{k,j(i)}$

•

$\theta =$ fixed effect slope coefficient for x_{ijklmp} – a continuous covariate for number of Statistics courses taken on the participant level

•

$\epsilon_{ijklmp} =$ residual error

- $\epsilon_{ijklmp} \overset{\text{iid}}{\sim} N(0, \sigma^2)$

◦

$\tau_m, \eta_{p(k,j(i))}, \epsilon_{ijklmp}$ all jointly independent

Levels for each fixed effect:

• Question ($\hat{\alpha}_i$)

– Question types 1, 2, 3, 4 (same as 5), 6 (same as 7), 8, 9, 10, 11, 12, 13

•

Assumption setting ($\beta_{j(i)}$)

– Question types 8, 12, 13 each had four unique underlying data-generating mechanisms, while the other question types had only two settings (assumptions met and not met)

-

Sample sizes (γ_k)

– Low and High

-

Loess setting (δ_ℓ)

– Yes and No

Here we justify the choice of fitting a linear mixed-effects model to a binary response. This strategy is an extension of the Linear Probability Model (LPM): using an ordinary linear model with a binary response (Caudill, 1988). While this approach is known to violate the assumptions of normality and constant variance (homoscedasticity), and may yield predictions outside the $(0,1)$ range (i.e., invalid probabilities), we argue that these issues are acceptable in our context. Specifically, because our primary interest lies in identifying general directions and obtaining point estimates of first-order trends in effects rather than conducting formal statistical inference, the standard errors and confidence intervals, which are most affected by these assumption violations, are of limited concern. Moreover, using a logistic or probit regression would yield results on a less interpretable scale, as well explained in Deke (2014), which discusses several advantages of an LPM in certain contexts:

Its main advantages are ease of implementation and interpretation. Specifically, the LPM can estimate impacts in cases where logistic regression cannot and, unlike logistic regression, the parameter estimates from the LPM can be directly interpreted as the impact of the intervention on the prevalence rate of the outcome.

Given our objective of exploring first-order effects rather than making precise probabilistic predictions, we find the linear model to be a more appropriate and interpretable choice.

3.3 Model Results

Table 2 and Table 3 present estimated coefficients for the mixed effects model, as outlined in the previous main section. Note that the intercept in this model, $\hat{\mu} = 0.539$, is an overall average rate of correct answers across the 7099 questions represented in this dataset. If respondents were selecting answers completely at random, the proportion of correct answers would be 0.470 (due to some questions having one correct option from among two choices, some having one correct option from among four choices, and some having two correct options from among four choices), so averaged across all participants and settings, students are only doing 7 percentage points better than random guessing at interpreting these plots.

The estimated question effects, $\hat{\alpha}_i$, were largely similar across levels, suggesting that the questions were of comparable difficulty. An exception was Question Type 8, which asked participants to identify which, if any, of response or predictor variable in a simple linear regression should be transformed. Particularly in the low sample size setting, this question proved challenging to most respondents, with many opting for both variables needing transformation, no matter the true data-generating mechanism.

Examining the assumption settings nested within questions, $\hat{\beta}_{j(i)}$, we see some noteworthy differences in plot assessment correctness depending on whether or not the assumptions were met in Questions 1, 2, 9, and 10. Questions 8 and 12 also have significant setting effect. In Question 1 (assessing normality based on histogram and quantile-quantile plot) participants were far more likely (a difference of 40 percentage points) to answer correctly when assumptions were met than when assumptions were not met, reflecting an overall preference for the answer "Yes", though in the low sample size the answer "No" was more common when assumptions were not met. For questions 2 (assessing symmetry), 9 (leverage), and 10 (Cook's distance), participants were more likely to answer correctly when the assumptions were not met, with estimated differences of approximately 20, 30, and 45 percentage points, respectively. For Question 8 (transformation of predictor and/or response), participants very much favored the option of transforming both variables, so this setting was the one in which the most correct answers were obtained. For Question 12 (testing equal means), participants struggled to give the correct answer (unequal-variance *t*-test) when the variances were unequal, often still selecting the equal-variance *t*-test

The estimated high sample size effect, $\hat{\gamma}_k$, was minimal and negative—approximately -0.3%, or -0.003—indicating that increasing sample size did not improve participants' ability to identify underlying distributional properties.

Adding loess curves to Question Types 4 (assessment of linearity) and 6 (assessment of constant variance) also had minimal impact (0.4%) on respondents' abilities to correctly identify non-linearity and non-constant variance, respectively. Looking at the plots in Figure A.5, however, we see that comparing question type 4 to Question Type 5, the proportion of correct responses was much larger when the generated data did not have a true linear relationship and a loess line was included. However, when there was a linear relationship, the loess line made participants possibly less likely to believe the relationship was linear.

Finally, the estimated slope for the number of statistics courses taken, $\hat{\theta}$, suggests a modest improvement in performance: each additional course was associated with an increase in accuracy of about 0.8%. However, this effect was not clearly discernible in Figure A.2.

Regarding the random effects, the estimated variability due to datasets, $\hat{\sigma}_\eta$, and due to participants, $\hat{\sigma}_\delta$, were approximately 8.4% and 4.4%, respectively. These provide useful context for interpreting the fixed effect estimates discussed above. It is noteworthy that the variability due to random dataset generation is roughly twice the variability between participants in the settings explored here, as in many previous studies, fixed datasets/plots were considered across all participants. The important and unavoidable element of random variability across realizations from the same data-generating mechanism has been acknowledged and underpins the approaches of tools like the `regressinator` and the simulated plot line-ups of Buja, Cook, Hofmann, Lawrence, Lee, Swayne and Wlckham (2009).

We also explored the impact of student status (undergraduate vs. graduate student) on correct answer rate. We did not include student status in the model, as it was highly correlated with the number of previous statistics classes taken, and we were more interested in the impact of number of statistics classes than the student status. However, the selection process inherent in becoming a graduate students could also contribute to an improved ability to interpret plots, so we separately investigated the rate of correct answers for graduate students (36) as compared to undergraduates (299) and non-student participants (5). We found that graduate students correctly interpreted an average of 62.7% (SD = 9.9%) of the plots they evaluated, which is notably higher than the average for undergraduates: 54.4% (SD = 14.2%). We also investigated whether there was a relationship between the number of questions answered and the rate of correctness, and found no evidence of a pattern in either direction.

To explore agreement across conditions, Pearson's chi-squared tests were applied to each subtable in Table 9. However, these *p*-values should not be interpreted as conventional significance tests; they serve merely as broad indicators of whether response patterns deviated from uniform randomness across cells.

4 Demonstration of Impact

As a demonstration of the impact of decisions based on diagnostic plot assessment, we consider Question 8, which presents a simple linear regression context and asks subjects to evaluate plots to choose whether the predictor and/or response variables should be transformed (via a simple log transformation in this example). This is a practice that is taught and recommended in introductory statistics courses at this and other institutions. The practice of choosing whether or not to transform variables on the basis of subjective plot assessments has several potential downstream consequences:

1) Inefficient or biased inferential performance

2) Lack of reproducibility: two analysts may well chose differently, even on the same dataset

3) Selection of a suboptimal model for answering the scientific question of interest and/or for making predictions of the response variable

For the data-generating settings presented to subjects in this question, the majority of viewers chose a "wrong" model—either transforming when unnecessary, or not transforming a variable when the true model did. Leaving aside the fact that in many real data settings we should not actually believe that any of our candidate models is the fundamental truth of the data-generating mechanism, we still must face the fact that this subjective choice inherently inhibits reproducible analyses and replicable findings, and does not directly optimize any quantifiable criteria.

When considering inferential performance, choice of transformation can affect propensity to reject null hypotheses/declare significance, and can also make our standard error estimates incorrect due to the presence of heteroskedasticity when different transformations are selected. To demonstrate this expected impact, we explored the data-generating mechanisms used in the paper, but simulated 10,000 datasets from each of the eight mechanisms:

• Two sample sizes:

– L: low = 30

– H: high = 100

•

Four possible transformation truths:

– B: both variables need transformation

– N: neither variable needs transformation

– X: only the predictor needs transformation

– Y: only the response needs transformation

For each of these 10,000 simulated datasets in each setting, we fit all four candidate models (Both transformed, Neither transformed, X transformed, Y transformed), and recorded the estimates, standard errors, and p-values for both the intercept and slope coefficients. We also calculated an approximation of the population mean squared prediction error (MSPE) for new observations on the untransformed response scale by generating a "population" dataset consisting of 100,000 observations and using the fitted model from each data-set to make predictions on these new observations.

In Tables 4 and 5, we examine the ability to correctly estimate the standard error of the intercept and slope, and find that (unsurprisingly) when an incorrect model is chosen, the standard error estimates are significantly biased compared to the true standard deviation of the sampling distributions. This is, of course, due to the non-constant variance under the choices of transformation that differ from the data-generating model scale.

We also explored, on each of the 10,000 simulated datasets for each of the eight data-generating settings, which of the fit settings produced the lowest MSPE on the new data. Interestingly, for the majority of simulated datasets in three of the four settings at both sample sizes, the model with the best prediction error was the model where neither variable was transformed. This is of course not a general finding, and is likely to be very different in different signal-to-noise ratio settings, for different coefficient values, and for different predictor distributions.

Finally, we can consider the MSPE of the chosen models for the specific datasets that were shown to survey participants in question 8. Again, we generate a large number (10,000,000) of new data observations from the same data-generating mechanism as was used to generate each of the eight original datasets to approximate the population, and then compare the MSPE for each of the four candidate models. In most settings, we see that the majority of respondents choose a model that gives less favorable predictions. While the differences in MSPE are generally not huge in this setting, there are in each setting a significant number of respondents who chose a model with much higher prediction error than the optimal choice.

5 Discussion

Valid statistical inference, as outlined in theory, often diverges markedly from how modeling is taught and applied in practice—even in many advanced courses. Students are commonly instructed to engage in exploratory data analysis by examining diagnostic plots, applying transformations based on visual cues, and refining models iteratively using hypothesis tests. They are told to then make inference from the final model as if it were pre-specified. While this process fosters an understanding of model selection strategies and promotes exploratory thinking, it overlooks a critical issue: the impact of model selection on statistical inference and the fact that it fundamentally changes the scientific questions addressed by the model. In particular, students are frequently taught to:

• Choose data transformations through guess-and-check procedures to satisfy model assumptions.

– This checking procedure often involves multiple hypothesis testing or some other form(s) of model selection.

•

Use diagnostic plots to guide model refinement without accounting for the inferential consequences of such decisions.

•

Interpret interaction or residual plots before conducting formal hypothesis tests—often even without adjusting for multiple comparisons (e.g., using Scheffé's method).

•

Drop highly correlated variables based on diagnostic tools (often solely based on pairwise correlation matrices and plots)

These practices—and many others like them—reinforce a flawed but pervasive workflow: performing exploratory analysis followed by confirmatory inference, without addressing the selection bias and subjectivity introduced in the process.

Statistical educators should emphasize that exploratory data analysis is not inherently problematic—rather, it becomes so when it is directly followed by inference without accounting for the preceding selection process. If these exploratory activities are of interest (as they often are and should be), instructors should clearly reiterate what sound practice requires: ideally, the pre-specification of an analysis plan prior to conducting any subsequent statistical inference. Going a step further, instructors could illustrate these consequences when teaching sampling distributions of estimators. Doing so would not only strengthen students—understanding of sampling distributions but also highlight how model selection uncertainty can meaningfully affect inference—an issue that arguably affects far more empirical research than is commonly recognized.

The survey highlights a central tension in applied statistical practice: even when faced with identical data and standard diagnostic plots, practitioners frequently diverge in how they assess assumptions and select modeling strategies. This variability persisted across training levels, sample sizes, and assumption settings, suggesting that subjective interpretation plays a larger role in model selection than often acknowledged in discussions of the final chosen analysis approach.

Such inconsistency is more than just noise—it has concrete implications for downstream inference. Each distinct modeling path effectively defines a different data-generating process, yet classical inference treats these decisions as fixed and correct. When model selection is informal or inconsistent, the resulting inferences can be misleading, overstated, or sensitive to unacknowledged sources of uncertainty.

Replicating this study but with much larger sample size, a broader population of participants, and more items could allow us to further explore questions such as the impact of the degree of assumption violation, plot characteristics, continuum of sample sizes, wording of question, subject academic background, and so on. This could quickly explode into far too many questions to succinctly report and summarize, and we expect that the general qualitative findings would be similar: people are inherently variable and imperfect in their subjective assessment of plots and modeling decisions. We are then left to assess the impact of this variability and imperfection in the analyses that are selected based on such practices.

6 Conclusion

The findings of this study underscore the fragility of workflows that rely on diagnostic interpretation without formal safeguards, and highlight the fact that the optimal analysis/modeling choice may differ depending upon the optimality criterion considered—but for all the considered criteria, human judgment based on plots was a poor selector of the modeling choice that gave the best results. This echoes broader concerns in post-model selection inference: variability in model choice—whether human or algorithmic—introduces uncertainty and potential bias that standard methods typically ignore. Addressing this requires methodological tools that go beyond point estimates and *p*-values to reflect the full uncertainty induced by the model selection process itself. The practice of using imprecise human visual judgments as a fundamental stage in an inferential process seems counter to the entire goal of a statistical analysis: a reproducible, objective, and precise quantified answer to a scientific question of interest.

7 Disclosure statement

The authors have no conflicts of interest to declare.

8 Data Availability Statement

Deidentified data and some example code have been made available at this URL:

https://doi.org/10.5281/zenodo.17346283

SUPPLEMENTARY MATERIALS

Example survey plots:

Survey plots from all settings for each question. (pdf file)

Survey Results Dataset:

Deidentified survey response data, linked above. (csv file)

Appendices

A.1 Survey Questions

1. Do you think the data in the histogram and quantile-quantile plot below could have plausibly come from a normal distribution?

2. The histograms below show a raw dataset (on the left) and the dataset after a log transformation (on the right). Do you think the distribution of the transformed data is approximately symmetric?

3. Based on the information below, would you consider performing variable selection to potentially remove one or more predictors from the model?

4. Some diagnostic plots for a multiple linear regression of Y vs. two predictors (X_1 and X_2) are shown below. Does the linearity assumption seem to hold?

5. The same as Question (4) but now with a LOESS curve overlaid.

6. A diagnostic plot for a simple linear regression of Y vs. X are shown below. Does the constant variance assumption seem to be satisfied?

7. The same as Question (6) but now with a LOESS curve overlaid.

8. Each column of plots below shows a simple linear regression and resulting diagnostic plots for different choices of transforming the response (Y); predictor (X) ; both (X and Y); or neither. Which set of plots seems to best fit the assumptions of a simple linear regression?

9. The plot below shows the leverage statistics for each observation in a simple linear regression. Based on this plot, do you think any of these points are outliers that should be investigated further and/or left out of the analysis?

10. The plot below shows the Cook's distance statistics for each observation in a simple linear regression. Based on this plot, do you think any of these points are outliers that should be investigated further and/or left out of the analysis?

11. The plot below shows the standardized residuals for each observation in a simple linear regression. Based on this plot, do you think any of these points are outliers that should be investigated further and/or left out of the analysis?

12. The plots below are some diagnostic plots to assess the assumptions for testing a hypothesis of equal means. Based on these plots, and assuming the independence assumption is met, which test would you suggest?

13. Forest plots, like the one shown below, are used to summarize results from multiple studies that are investigating the same interventions or comparisons. For each study, a confidence interval for the true effect size based on the data for that study is shown. Based on the information from all the studies shown, do you think it is reasonable to conclude that the true treatment effect across all these studies is non-zero?

A.2 Survey Answers

A.3 Figures and Tables

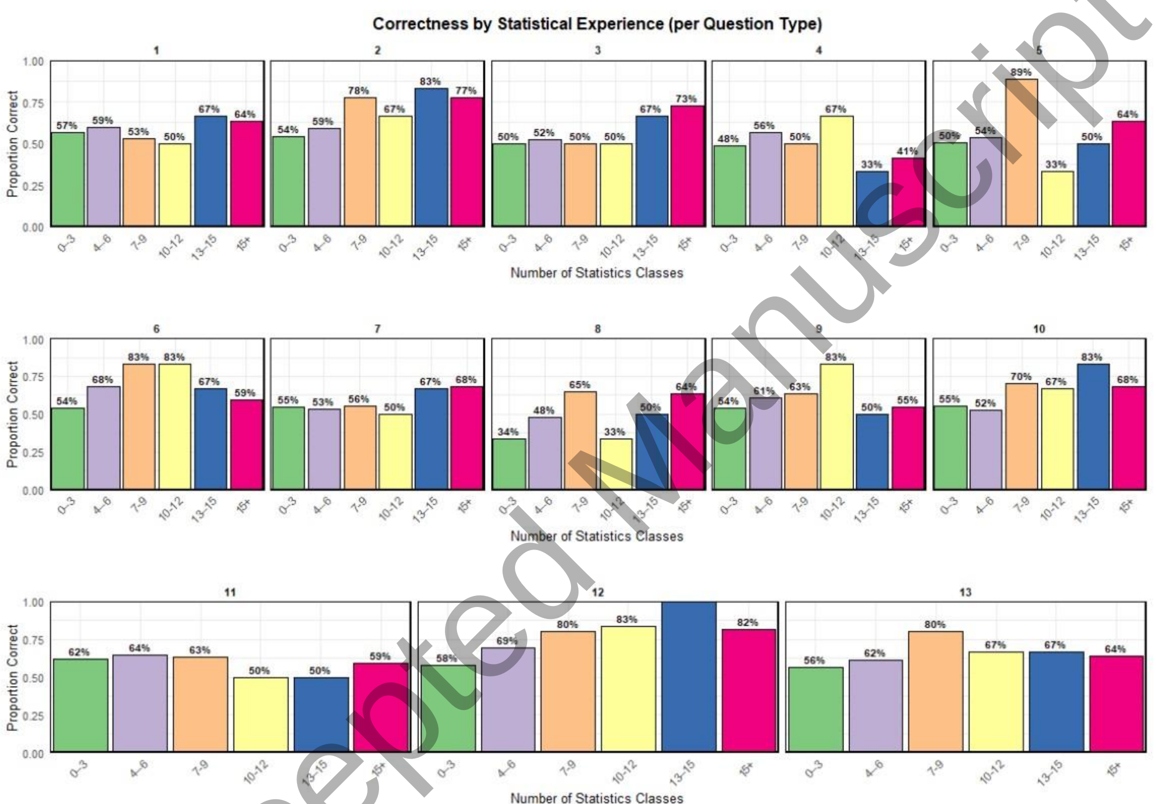

Figure A.2: Proportion Correct Across All Questions and Subjects - Investigating Effect of Statistical Education

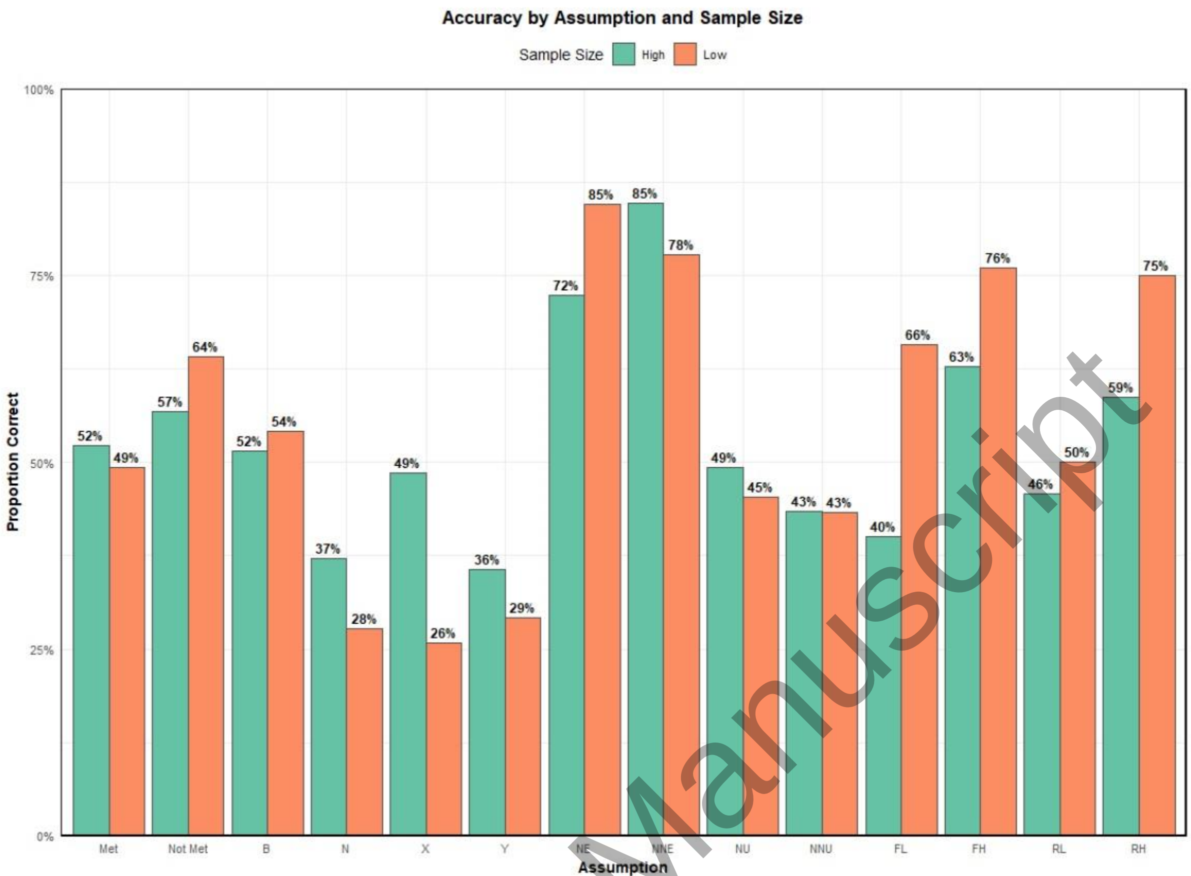

Figure A.3: Proportion Correct Across All Questions and Subjects - Comparing Sample Size Across Different Assumption Settings

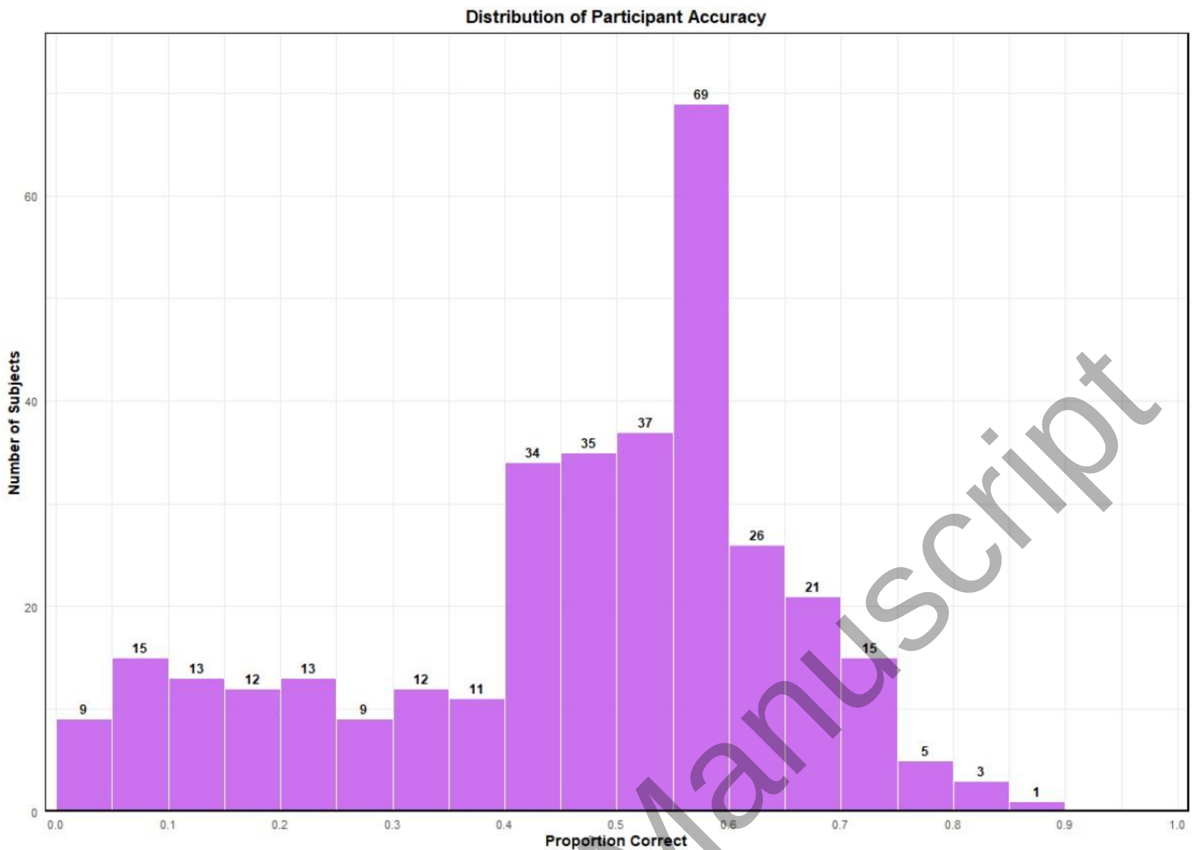

Figure A.4: Number of Correct Answers Frequencies Among All Participants

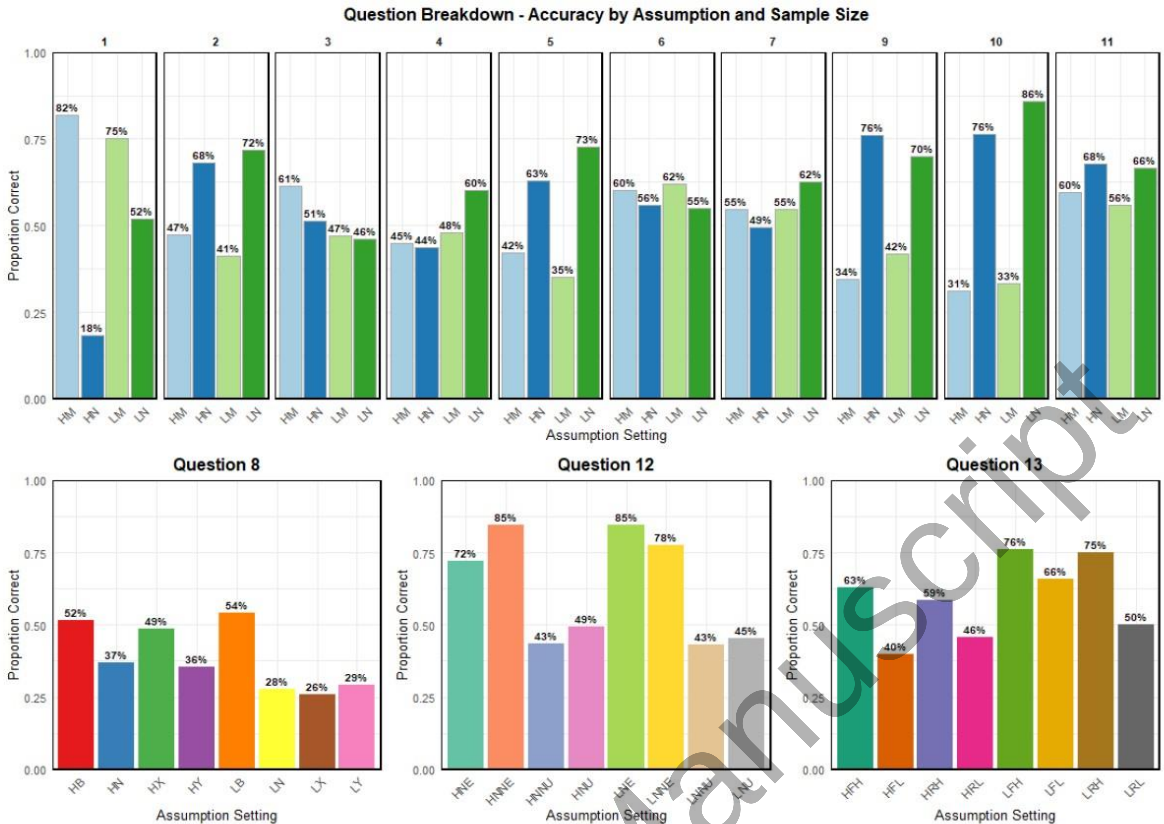

Figure A.5: For a Given Question Type: Answer Correctness by Assumption Settings

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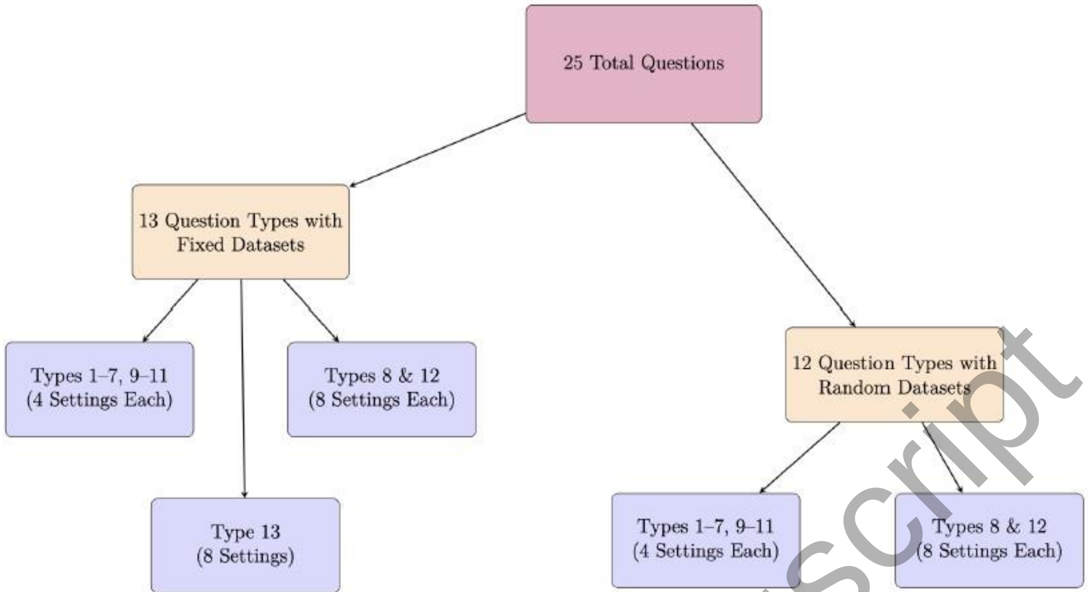

Figure 1: Structure of the 25-question survey, organized by dataset type and question categories. Given in a random order, fixed questions were shown identically to all participants, while random-dataset questions varied by respondent. Each branch includes different combinations of question types and settings, as described below.

Table 1: **Proportion Correct Answers by Question Types. Question Type 12 abbreviations:** N = Normally distributed; NN = Non-normally distributed; E = Equal variance; U = Unequal variance. **Question Type 13 abbreviations:** Fix = studies had the same fixed treatment effect; Rand = studies had random treatment effects; Zero = true population mean treatment effect was zero; Non-zero = true population mean treatment effect was non-zero.

Question Type	**Proportion Correct Answers (Percentage Correct)**			
	Sample Size, Assumption Setting			
	H, Met	**H, Not Met**	**L, Met**	**L, Not Met**
1 (Q1, Q13) Normality	122/149 (81.9%)	26/139 (18.7%)	110/146 (75.3%)	71/137 (51.8%)
2 (Q2, Q14) Symmetry	74/153 (48.4%)	104/156 (66.7%)	53/129 (41.1%)	99/139 (71.2%)
3 (Q3, Q15) Multicollinearity	90/146 (61.6%)	74/143 (51.8%)	68/142 (47.9%)	68/150 (45.3%)
4 (Q4, Q16) Linearity	58/130 (44.6%)	61/140 (43.6%)	72/151 (47.7%)	90/149 (60.4%)
5 (Q5, Q17) Linearity (w/ LOESS)	67/159 (42.1%)	83/132 (62.9%)	49/140 (35.0%)	100/137 (73.0%)
6 (Q6, Q18) Constant Variance	93/154 (60.4%)	83/148 (56.1%)	86/140 (61.4%)	72/132 (54.6%)
7 (Q7, Q19) Constant Variance (w/ LOESS)	76/139 (54.7%)	74/152 (48.7%)	82/149 (55.0%)	85/136 (62.5%)
9 (Q9, Q21) Leverage	43/126 (34.1%)	96/126 (76.2%)	72/171 (42.1%)	115/163 (70.6%)
10 (Q10, Q22) Cook’s Distance	44/146 (30.1%)	108/141 (76.6%)	49/149 (32.9%)	120/140 (85.7%)
11 (Q11, Q23) Standardized Residuals	98/164 (59.8%)	94/138 (68.1%)	75/134 (56.0%)	91/136 (66.9%)
8 (Q8, Q20) Transformation	**H, Both**	**H, Neither**	**H, X**	**H, Y**
	34/66 (51.5%)	33/89 (37.1%)	34/70 (48.6%)	21/61 (34.4%)
	L, Both	**L, Neither**	**L, X**	**L, Y**
	40/73 (54.8%)	19/66 (28.8%)	16/62 (25.8%)	26/89 (29.2%)

12 (Q12, Q24) Hypothesis Test	**H, N, E**	**H, NN, E**	**H, NN, U**	**H, N, U**
	47/65 (72.3%)	56/66 (84.8%)	31/70 (44.3%)	34/69 (49.3%)
	L, N, E	**L, NN, E**	**L, NN, U**	**L, N, U**
	71/84 (84.5%)	49/63 (77.8%)	33/76 (43.4%)	34/75 (45.3%)
13 (Q25) Forest Plot	**H, Fix, Non-zero**	**H, Fix, Zero**	**H, Rand, Non-zero**	**H, Rand, Zero**
	22/35 (62.9%)	13/31 (41.9%)	27/46 (58.7%)	16/35 (45.7%)
	L, Fix, Non-zero	**L, Fix, Zero**	**L, Rand, Non-zero**	**L, Rand, Zero**
	20/26 (76.9%)	23/35 (65.7%)	27/36 (75.0%)	21/43 (48.8%)

Table 2: **Survey Model Coefficient Estimates (Part 1)**

Effect		Coefficient
Intercept ($\hat{\mu}$)		0.539
Sample Size ($\hat{\gamma}_k$)	Sample Size (High)	-0.003
	Sample Size (Low)	0.003
Loess ($\hat{\delta}_\ell$)	Loess Curves (Yes)	0.004
	Loess Curves (No)	-0.004
Number of Stat Courses ($\hat{\theta}$)		0.008
Dataset ($\hat{\sigma}_\eta$)		0.084
Participant ($\hat{\sigma}_\tau$)		0.044
Residual Error ($\hat{\sigma}_\epsilon$)		0.469

Table 3: **Survey Model Coefficient Estimates (Part 2)**

Effect		Coefficient
Questions ($\hat{\alpha}_i$)	Question Type 1	0.002
	Question Type 2	0.054
	Question Type 3	-0.054
	Question Type 4	-0.029
	Question Type 6	-0.001
	Question Type 8	-0.173
	Question Type 9	0.010
	Question Type 10	0.010
	Question Type 11	0.073
	Question Type 12	0.076
	Question Type 13	0.032
Settings ($\hat{\beta}_{j(i)}$)	Question Type 1 Setting (Met)	0.209
	Question Type 1 Setting (Not Met)	-0.209
	Question Type 2 Setting (Met)	-0.093
	Question Type 2 Setting (Not Met)	0.093
	Question Type 3 Setting (Met)	0.025
	Question Type 3 Setting (Not Met)	-0.025
	Question Type 4 Setting (Met)	-0.056
	Question Type 4 Setting (Not Met)	0.056
	Question Type 6 Setting (Met)	0.017
	Question Type 6 Setting (Not Met)	-0.017
	Question Type 8 Setting (Both)	0.139

Effect		Coefficient
	Question Type 8 Setting (Neither)	-0.064
	Question Type 8 Setting (X)	-0.006
	Question Type 8 Setting (Y)	-0.069
	Question Type 9 Setting (Met)	-0.158
	Question Type 9 Setting (Not Met)	0.158
	Question Type 10 Setting (Met)	-0.233
	Question Type 10 Setting (Not Met)	0.233
	Question Type 11 Setting (Met)	-0.048
	Question Type 11 Setting (Not Met)	0.048
	Question Type 12 Setting (NE)	0.141
	Question Type 12 Setting (NNE)	0.193
	Question Type 12 Setting (NU)	-0.153
	Question Type 12 Setting (NNU)	-0.181
	Question Type 13 Setting (FH)	0.095
	Question Type 13 Setting (RH)	0.066
	Question Type 13 Setting (FL)	-0.050
	Question Type 13 Setting (RL)	-0.111

Table 4: Ratio of (average estimated SE across datasets) to (SD of estimates across datasets) for Intercept (β_0). Values closest to 1 are bold, indicating that the estimated standard errors are approximating the true sampling distribution well; values far from 1 indicate that the model is doing a poor job estimating the variability of its estimates.

	Fit Setting			
	Both X and Y	**Neither**	**X**	**Y**
True Setting	**transformed**	**transformed**	**transformed**	**transformed**
Low, Both	**0.986**	0.199	0.283	0.903
Low, Neither	0.865	**0.979**	0.871	0.890
Low, X	0.989	0.844	**1.001**	0.793
Low, Y	0.960	0.160	0.174	**0.974**
High, Both	**0.991**	0.353	0.525	0.875
High, Neither	0.863	**1.008**	0.858	0.905
High, X	0.987	0.759	**0.993**	0.744
High, Y	0.979	0.216	0.261	**1.000**

Table 5: Ratio of (average estimated SE across datasets) to (SD of estimates across datasets) for Slope (β_1). Values closest to 1 are bold, indicating that the estimated standard errors are approximating the true sampling distribution well; values far from 1 indicate that the model is doing a poor job estimating the variability of its estimates.

	Fit Setting			
	Both X and Y	**Neither**	**X**	**Y**
True Setting	**transformed**	**transformed**	**transformed**	**transformed**
Low, Both	**0.980**	0.094	0.163	0.756
Low, Neither	0.879	**0.978**	0.873	0.920
Low, X	0.867	0.645	**0.982**	0.669
Low, Y	0.962	0.145	0.163	**0.975**
High, Both	**0.993**	0.178	0.243	0.685
High, Neither	0.879	**1.009**	0.860	0.942
High, X	0.875	0.551	**0.995**	0.586
High, Y	0.980	0.196	0.243	**1.000**

Table 6: Number of times each fit setting achieved the lowest MSPE across 10,000 simulated datasets

	Fit Setting			
	Both X and Y	**Neither**	**X**	**Y**
True Setting	**transformed**	**transformed**	**transformed**	**transformed**
Low, Both	1350	**5666**	2977	7
Low, Neither	2209	**6334**	101	1356
Low, X	2723	26	**7248**	3
Low, Y	304	**7133**	1023	1540
High, Both	817	**7384**	1799	0
High, Neither	1400	**8278**	0	322
High, X	1837	0	**8163**	0
High, Y	2	**8737**	288	973

Table 7: Number of respondents selecting each model for a given setting (with average MSPE on new data for that model in parentheses).

	Selected Response (MSPE)			
	Both X and Y	**Neither**	**X**	**Y**
True Setting	**transformed**	**transformed**	**transformed**	**transformed**
Low, Both	39 (2.44×10^{3})	7 ($\mathbf{2.40\times10^{3}}$)	15 (2.42×10^{3})	11 (1.42×10^{68})
Low, Neither	18 ($\mathbf{4.20\times10^{0}}$)	18 (5.15×10^{0})	9 (4.41×10^{0})	20 (2.78×10^{7})
Low, X	28 ($\mathbf{4.20\times10^{0}}$)	6 (5.15×10^{0})	16 (4.41×10^{0})	12 (2.78×10^{7})
Low, Y	38 (7.70×10^{7})	16 ($\mathbf{7.65\times10^{7}}$)	9 (7.66×10^{7})	26 (7.66×10^{7})
High, Both	34 (1.90×10^{3})	9 ($\mathbf{1.89\times10^{3}}$)	8 (1.90×10^{3})	15 (3.01×10^{100})
High, Neither	31 (4.25×10^{0})	33 ($\mathbf{4.13\times10^{0}}$)	12 (4.21×10^{0})	13 (4.23×10^{9})
High, X	18 (4.12×10^{0})	5 (5.29×10^{0})	33 ($\mathbf{4.07\times10^{0}}$)	12 (6.50×10^{8})
High, Y	28 (5.69×10^{7})	3 (5.72×10^{7})	7 (5.70×10^{7})	21 ($\mathbf{5.68\times10^{7}}$)

Table 8: **Survey Responses Summary**

Question Type	Assumption / Setting	Correct Answer
1 (Q1, Q13)	M	Yes
	N	No
2 (Q2, Q14)	M	Yes
	N	No
3 (Q3, Q15)	M	No
	N	Yes
4 (Q4, Q16)	M	Yes
	N	No
5 (Q5, Q17)	M	Yes
	N	No
6 (Q6, Q18)	M	Yes
	N	No
7 (Q7, Q19)	M	Yes
	N	No
8 (Q8, Q20)	B	Both Transformed
	N	Neither Transformed
	X	X Transformed
	Y	Y Transformed
9 (Q9, Q21)	M	No
	N	Yes
10 (Q10, Q22)	M	No
	N	Yes

11 (Q11, Q23)	M	No
	N	Yes
12 (Q12, Q24)	NE	Equal or Unequal Variance *t*-test
	NNE	Equal or Unequal Variance *t*-test
	NNU	Unequal Variance *t*-test
	NU	Unequal Variance *t*-test
13 (Q25)	H	Yes
	L	No

Table 9: Survey Responses by Question Type and Sampling Assumption. Each column label encodes the sampling scenario and the correct answer. The first letter in each column indicates the sample size: H = High, L = Low. For **Questions 1–7, 9-11** (Yes/No format), the second letter represents the correct answer (Y = Yes, N = No). For **Question 8**, the second letter indicates which variable should be transformed: B = Both, N = Neither, X = Predictor, Y = Response. For **Question 12**, the second and third letters reflect data assumptions: N = Normality met, NN = Normality not met; E = Equal variance, U = Unequal variance. For **Question 13**, the second letter denotes the model type (F = Fixed effects, R = Random effects), and the third letter indicates study count (H = High, L = Low). This coding scheme facilitates comparison of survey responses across different statistical scenarios.

Q1: Normality Assessment

	HY	HN	LY	LN
No	27	26	36	71
Yes	122	113	110	66

$\chi^2_{(3)} = 52.529, p = 2.311 \times 10^{-11}$

Q2: Symmetry Assessment

	HY	HN	LY	LN
No	79	105	76	99
Yes	74	51	53	40

$\chi^2_{(3)} = 14.404, p = 0.00240$

Q3: Multicollinearity Assessment

	HN	HY	LN	LY
No	90	69	68	82
Yes	56	74	74	68

$\chi^2_{(3)} = 7.3263, p = 0.06219$

Q4: Linearity Assessment

	HY	HN	LY	LN
No	72	61	79	90
Yes	58	79	72	59

$\chi^2_{(3)} = 8.5987, p = 0.03513$

Q5: Linearity Assessment (w/ LOESS Curve)

	HY	HN	LY	LN
No	92	83	91	100
Yes	67	49	49	37

$\chi^2_{(3)} = 7.5353, p = 0.0566$

Q6: Constant Variance Assessment

	HY	HN	LY	LN
No	61	83	54	72
Yes	93	65	86	60

$\chi^2_{(3)} = 15.283, p = 0.00159$

Q7: Constant Variance Assessment (w/ LOESS Curve)

	HY	HN	LY	LN
No	63	74	67	85
Yes	76	78	82	51

$\chi^2_{(3)} = 11.13, p = 0.01104$

Q9: Case Influence Statistic Leverage Assessment

	HN	HY	LN	LY
No	43	30	72	48
Yes	83	96	99	115

$\chi^2_{(3)} = 12.239, p = 0.0066$

Q8: Transformation Decision

	HB	HN	HX	HY	LB	LN	LX	LY
Both	34	31	19	30	40	18	28	38
None	9	33	5	3	7	19	6	16
X	8	12	34	7	15	9	16	9
Y	15	13	12	21	11	20	12	26

$\chi^2_{(21)} = 110.01, p = 4.7 \times 10^{-14}$

Q10: Case Influence Statistic Cook's Distance Assessment

	HN	HY	LN	LY
No	44	33	49	20
Yes	102	108	100	120

$\chi^2_{(3)} = 15.579, p = 0.00138$

Q11: Case Influence Statistic Standardized Residuals Assessment

	HN	HY	LN	LY
No	98	44	75	45
Yes	66	94	59	91

$\chi^2_{(3)} = 37.448, p = 3.699 \times 10^{-8}$

Q12: Hypothesis Test Decision

	HNE	HNNE	HNNU	HNU	LNE	LNNE	LNNU	LNU
EV-t	22	23	26	24	37	20	25	26
Perm	3	4	2	3	3	7	7	5
Welch	25	33	31	34	34	29	33	34
WRS	15	6	11	8	10	7	11	10

$\chi^2_{(21)} = 16.717, p = 0.7281$

Q13: Forest Plot Evaluation

	HFH	HFL	HRH	HRL	LFH	LFL	LRH	LRL
No	13	13	19	16	6	23	9	21
Yes	22	18	27	19	20	12	27	22

$\chi^2_{(7)} = 17.562, p = 0.01411$